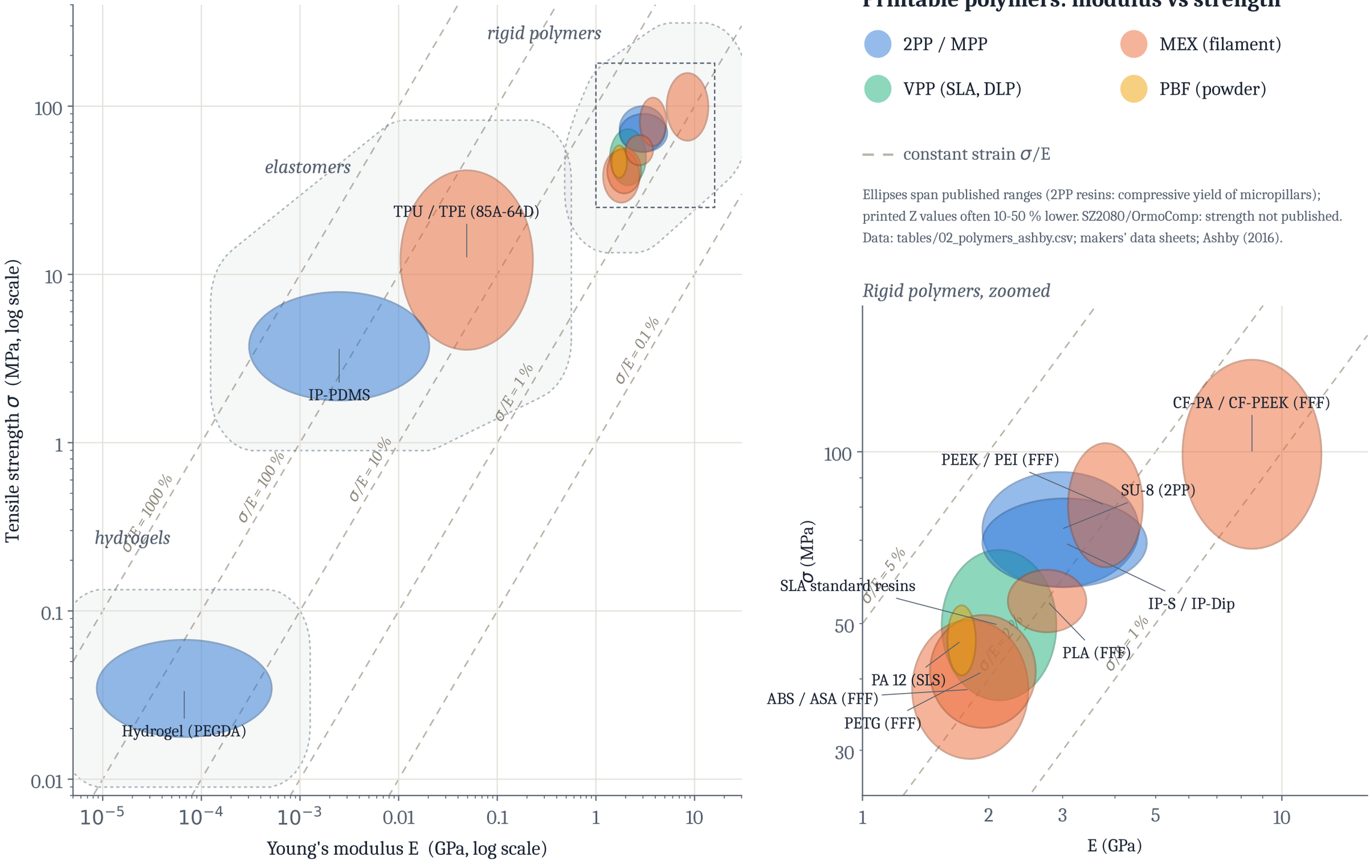

Printable polymers: modulus vs strength
2PP / MPP
MEX (filament)
VPP (SLA, DLP)
PBF (powder)
constant strain σ/E
Ellipses span published ranges (2PP resins: compressive yield of micropillars); printed Z values often 10-50 % lower. SZ2080/OrmoComp: strength not published.
Data: tables/02_polymers_ashby.csv; makers' data sheets; Ashby (2016).
rigid polymers
elastomers
hydrogels
TPU / TPE (85A-64D)
IP-PDMS
Hydrogel (PEGDA)
σ/E = 0.1 %
σ/E = 1 %
σ/E = 10 %
σ/E = 100 %
σ/E = 1000 %
Tensile strength σ (MPa, log scale)
Young's modulus E (GPa, log scale)
Rigid polymers, zoomed
CF-PA / CF-PEEK (FFF)
PEEK / PEI (FFF)
SU-8 (2PP)
IP-S / IP-Dip
SLA standard resins
PLA (FFF)
PA 12 (SLS)
ABS / ASA (FFF)
PETG (FFF)
σ/E = 5 %
σ/E = 2 %
σ/E = 1 %
σ (MPa)
E (GPa)